

worse. Why? Clearly, catastrophic events will be necessarily absent from the data, since the survivorship of the variable itself will depend on such effect. Thus such distributions will let the observer become prone to overestimation of stability and underestimation of potential volatility and risk.

This point—that things have a bias to appear more stable and less risky in the past, leading us to surprises—needs to be taken seriously, particularly in the medical field. The history of epidemics, narrowly studied, does not suggest the risks of the great plague to come that will dominate the planet. Also I am convinced that in doing what we are to the environment, we greatly underestimate the potential instability we will experience somewhere from the cumulative damage we have done to nature.

One illustration of this point is playing out just now. At the time of writing, the stock market has proved much, much riskier than innocent retirees were led to believe from historical discourses showing a hundred years of data. It is down close to 23 percent for the decade ending in 2010, while the retirees were told by finance charlatans that it was expected to rise by around 75 percent over that time span. This has bankrupted many pension plans (and the largest car company in the world), for they truly bought into that “empirical” story—and of course it has caused many disappointed people to delay their retirement. Consider that we are suckers and will gravitate toward *those variables that are unstable but that appear stable*.

Preasymptotics. Let us return to Platonicity with a discussion of preasymptotics, what happens in the short term. Theories are, of course, a bad thing to start with, but they can be worse in some situations when they were derived in idealized situations, the asymptote, but are used outside the asymptote (its limit, say infinity or the infinitesimal). Mandelbrot and I showed how some asymptotic properties do work well preasymptotically in Mediocristan, which is why casinos do well; matters are different in Extremistan.

Most statistical education is based on these asymptotic, Platonic properties, yet we live in the real world, which rarely resembles the asymptote. Statistical theorists know it, or claim to know it, but not your regular user of statistics who talks about “evidence” while writing papers. Furthermore, this compounds what I called the ludic fallacy: most of what students of mathematical statistics do is assume a structure similar to the closed structures of games, typically with a priori known probability. Yet the problem we have is not so much making computations once you know the probabilities, but finding the true